

Chapter 8

Vision-Language Models

Exam Study Guide, Worked Concepts, Sample Questions, and

Prepared from	uploaded course materials
Main source	Chapter_8_Vision_Language_Models.ipynb
Exam source	Chapter_8_Summary_and_Sample_Questins.pdf
Practice source	Chapter_8_Flashcards.pdf

How to use it

Study VLMs by separating alignment models from fusion/generative models. CLIP-style models learn shared embeddings; encoder-decoder models use cross-attention for grounded generation.

Study order: First read the concept sections, then solve the five sample exam questions, then test yourself with the flashcards.

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- 1. Quick exam map
- 2. What VLMs learn
- 3. Dual-encoder contrastive models
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Most useful mental shortcut
VLMs either align image and text in a shared embedding space or fuse visual tokens into a generative text decoder.

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1. Quick exam map

Vision-language models connect visual meaning and linguistic meaning so images and text can be compared, retrieved, or generated together.

Image

visual input

Image encoder

visual embedding or tokens

Text

caption, label, or prompt

Text

encoder/decoder language embedding or generation

Alignment/fusion

matching, retrieval, captioning, VQA Figure: the main method pipeline for this chapter.

Exam topic	What you must be able to say
Goal of VLM	Connect visual meaning and linguistic meaning.

Exam topic	What you must be able to say
Dual encoder	Separate image/text encoders trained by contrastive learning.
CLIP objective	Pull matched pairs together and push mismatched pairs apart.
Zero-shot classification	Compare image embedding to prompt embeddings.
Fusion VLM	Text decoder cross-attends to visual memory for generation.
Failure modes	Bias, weak grounding, distribution shift, hallucination.

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2. Zero-prior explanation: understand the

idea before the equations This section assumes you have not read the lecture yet. The goal is to build intuition first, then connect the intuition to the equations and exam language.

What VLMs learn

A VLM should know that an image, a caption, and a phrase can refer to the same concept even though they are different modalities.

Dual encoders

CLIP-like models encode image and text separately into a shared embedding space. They are efficient because embeddings can be precomputed and compared by cosine similarity.

Contrastive learning

A batch contains matched image-text pairs. Correct pairs are positives; mismatched pairs are negatives. This teaches the model not only what matches, but also what does not match.

Memory line
Alignment is global matching; fusion is token-level conditioning.

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3. Core concepts and worked explanations

Zero-shot classification

Instead of a fixed classifier, write text prompts for each class and compare the image embedding to the prompt embeddings.

Fusion models

Fusion VLMs condition a text decoder on visual tokens through cross-attention. This supports captions, VQA, and grounded generation.

Limitations

Global alignment can miss local details. Generative VLMs can hallucinate plausible text that is not supported by the image. Prompt wording, bias, and domain shift matter.

4. Method explanation, worked example,

and application

Step	Method explanation
Encode modalities	Represent image and text using encoders or fusion architecture.
Align or fuse	Dual encoders compare embeddings; fusion models cross-attend.
Train objective	Use contrastive loss or autoregressive loss.
Perform task	Retrieve, classify zero-shot, answer, or caption.

Worked example
Example: for zero-shot classification, embed an image and prompts such as "a photo of a cat" and "a photo of a dog". The label with highest cosine similarity becomes the predicted class.

Applications and usage
Usage: VLMs support image-text retrieval, zero-shot recognition, captioning, visual question answering, grounding, document understanding, and multimodal chat.

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5. Formula / definition cheat sheet

Read formulas as compact sentences. Do not memorize symbols blindly; use the symbol key table after the formula table to understand what each part represents.

Formula	Interpretation
$v=f_img(image), t=f_txt(text)$ Image and text embeddings.	$s(image, text)=v^T t / \tau$ Temperature-scaled similarity.
$P(text_i image_i)=\exp(s_{ii})/\sum_j \exp(s_{ij})$ Contrastive image-to-text probability.	$L_{CLIP}=(L_{image \rightarrow text}+L_{text \rightarrow image})/2$ Symmetric contrastive loss.
$p(y image)=\prod_t p(y_t y_{<t}, image)$ Generative caption/VQA factorization.	$CrossAttn(Q_text, K_img, V_img)$ Fusion mechanism.

6. Equation symbol key table

This table explains the main symbols used in the equations. When studying, cover the meaning column and check whether you can explain each symbol clearly.

Symbol	Meaning	Plain-English reading
v	Image embedding	Normalized vector from image encoder.

Symbol	Meaning	Plain-English reading
t	Text embedding	Normalized vector from text encoder.
τ	Temperature	Controls sharpness of similarity
probabilities.	s_{ij}	Similarity score
Compatibility between image i and text	$j.$	y
Generated text	Caption or answer token sequence.	Q_{text}
Text queries	Decoder tokens asking visual memory	for information.
$K_{\text{img}}, V_{\text{img}}$	Visual keys/values	Image features used by cross-attention.

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7. Main code prompt / implementation

skeleton This is not meant to replace the lecture code. It is a compact prompt showing the computational pattern you should recognize in homework and exams.

```
# CLIP-like zero-shot prompt
image_emb = normalize(image_encoder(image)) text_embs = normalize(text_encoder(class_prompts)) scores = image_emb
# Fusion captioning prompt

visual_tokens = image_encoder(image) caption = decoder.generate(prompt,
cross_attention_memory=visual_tokens)
```

How to explain the code in words
 Start by identifying the input, the model output, the objective being minimized, and the update rule.
 Then connect the code back to the formulas and the method table.

8. Common exam traps

Trap	Correct way to think
Thinking CLIP generates	captions CLIP scores compatibility; generation needs a decoder.
Training only positive pairs	Contrastive learning needs negatives.
Equating global alignment with	grounding Global embeddings can miss local details.

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9. Exam-style sample questions with

answers These are written in the style of conceptual exam answers. Use them to practice explaining, not only recognizing terms.

Question 1

Why does CLIP need negative pairs?

Sample answer

Without mismatches, embeddings can collapse and fail to separate concepts.

Question 2

Why does zero-shot classification work?

Sample answer

Class names enter through text prompts in the shared embedding space.

Question 3

Why is CLIP good for retrieval?

Sample answer

Image and text embeddings are computed independently and compared efficiently.

Question 4

Why can fusion be better for grounding?

Sample answer

Text tokens can cross-attend to visual tokens for localized evidence.

Question 5

What is hallucination in VLMs?

Sample answer

Generating plausible text not supported by the image.

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10. Flashcards for rapid review

Use these for active recall. Cover the answer column and try to answer before checking.

Card 1

Q: What is a VLM?

A: A model connecting image and language meaning.

Card 2

Q: What is CLIP?

A: A contrastive dual-encoder VLM.

Card 3

Q: What does L2 normalization enable?

A: Cosine similarity by dot product.

Card 4

Q: What is zero-shot classification?

A: Classify using text prompts.

Card 5

Q: Does CLIP generate captions?

A: Not by itself.

Card 6

Q: What is cross-attention?

A: Text attending to visual memory.

Card 7

Q: What is hallucination?

A: Unsupported generated content.

Card 8

Q: What is retrieval?

A: Ranking matches in embedding space.

Card 9

Q: What must you remember about Goal of VLM?

A: Connect visual meaning and linguistic meaning.

Card 10

Q: What must you remember about Dual encoder?

A: Separate image/text encoders trained by contrastive learning.

Card 11

Q: What must you remember about CLIP objective?

A: Pull matched pairs together and push mismatched pairs apart.

Card 12

Q: What must you remember about Zero-shot classification?

A: Compare image embedding to prompt embeddings.

Card 13

Q: What must you remember about Fusion VLM?

A: Text decoder cross-attends to visual memory for generation.

Card 14

Q: What must you remember about Failure modes?

A: Bias, weak grounding, distribution shift, hallucination.

Card 15

Q: What does this formula represent:

$v=f_{\text{img}}(\text{image}), t=f_{\text{txt}}(\text{text})$?

A: Image and text embeddings.

Card 16

Q: What does this formula represent:

$$s(\text{image}, \text{text}) = v^T t / \tau$$

A: Temperature-scaled similarity.

Card 17

Q: What does this formula represent:

$$P(\text{text}_i | \text{image}_i) = \exp(s_{ii}) / \sum_j \exp(s_{ij})$$

A: Contrastive image-to-text probability.

Card 18

Q: What does this formula represent:

$$\text{LCLIP} = (\text{L}_{\text{image} \rightarrow \text{text}} + \text{L}_{\text{text} \rightarrow \text{image}}) / 2$$

A: Symmetric contrastive loss.

Card 19

Q: What does this formula represent:

$$p(y | \text{image}) = \prod_t p(y_t | y_{< t}, \text{image})$$

A: Generative caption/VQA factorization.

Card 20

Q: What does this formula represent:

$$\text{CrossAttn}(Q_{\text{text}}, K_{\text{img}}, V_{\text{img}})$$

A: Fusion mechanism.

Card 21

Q: What does v mean?

A: Image embedding. Normalized vector from image encoder.

Card 22

Q: What does t mean?

A: Text embedding. Normalized vector from text encoder.

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Card 23

Q: What does τ mean?

A: Temperature. Controls sharpness of similarity probabilities.

Card 24

Q: What does s_{ij} mean?

A: Similarity score. Compatibility between image i and text j .

Card 25

Q: What does y mean?

A: Generated text. Caption or answer token sequence.

Card 26

Q: What does Q_{text} mean?

A: Text queries. Decoder tokens asking visual memory for information.

Card 27

Q: What does $K_{\text{img}}, V_{\text{img}}$ mean?

A: Visual keys/values. Image features used by cross-attention.

Card 28

Q: What is the key idea of What VLMs learn?

A: A VLM should know that an image, a caption, and a phrase can refer to the same concept even though they are different

Card 29

Q: What is the key idea of Dual encoders?

A: CLIP-like models encode image and text separately into a shared embedding space. They are efficient because embeddings can be precomputed and compared by cosine

Card 30

Q: What is the key idea of Contrastive learning?

A: A batch contains matched image-text pairs.

Correct pairs are positives; mismatched pairs are negatives. This teaches the model not only what matches, but also what does not

Card 31

Q: What is the key idea of Zero-shot classification?

A: Instead of a fixed classifier, write text prompts for each class and compare the image embedding to the prompt

Card 32

Q: What is the key idea of Fusion models?

A: Fusion VLMs condition a text decoder on visual tokens through cross-attention. This supports captions, VQA, and grounded

Card 33

Q: What is the key idea of Limitations?

A: Global alignment can miss local details.

Generative VLMs can hallucinate plausible text that is not supported by the image. Prompt wording, bias, and domain shift

Card 34

Q: What happens in the step: Encode modalities?

A: Represent image and text using encoders or fusion architecture.

Card 35

Q: What happens in the step: Align or fuse?

A: Dual encoders compare embeddings; fusion models cross-attend.

Card 36

Q: What happens in the step: Train objective?

A: Use contrastive loss or autoregressive loss.

Card 37

Q: What happens in the step: Perform task?

A: Retrieve, classify zero-shot, answer, or caption.

Card 38

Q: Can you separate alignment and fusion?

A: Shared embedding vs cross-attended generation.

Card 39

Q: Can you explain CLIP positives/negatives?

A: Matched pairs close, mismatches far.

Card 40

Q: Can you explain zero-shot?

A: Compare image to class prompts.

Card 41

Q: Can you name VLM failure modes?

A: Bias, weak grounding, shift, hallucination.

Card 42

Q: Why does CLIP need negative pairs?

A: Without mismatches, embeddings can collapse and fail to separate concepts.

Card 43

Q: Why does zero-shot classification work?

A: Class names enter through text prompts in the shared embedding space.

Card 44

Q: Why is CLIP good for retrieval?

A: Image and text embeddings are computed independently and compared efficiently.

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Card 45

Q: Why can fusion be better for grounding?

A: Text tokens can cross-attend to visual tokens for localized evidence.

Card 46

Q: What is hallucination in VLMs?

A: Generating plausible text not supported by the image.

Card 47

Q: In plain English, why is $v=f_{\text{img}}(\text{image})$, $t=f_{\text{txt}}(\text{text})$ important?

A: It captures: Image and text embeddings. Use it as a compact exam sentence, then define each symbol.

Card 48

Q: In plain English, why is $s(\text{image}, \text{text}) = v^T t / \tau$ important?

A: It captures: Temperature-scaled similarity. Use it as a compact exam sentence, then define each symbol.

Card 49

Q: In plain English, why is $P(\text{text}_i | \text{image}_i) = \exp(s_{ii}) / \sum_j \exp(s_{ij})$ important?

A: It captures: Contrastive image-to-text probability. Use it as a compact exam sentence, then define each symbol.

Card 50

Q: In plain English, why is $L_{\text{CLIP}} = (L_{\text{image} \rightarrow \text{text}} + L_{\text{text} \rightarrow \text{image}}) / 2$ important?

A: It captures: Symmetric contrastive loss. Use it as a compact exam sentence, then define each symbol.

Card 51

Q: In plain English, why is $p(y | \text{image}) = \prod_t p(y_t | y_{<t}, \text{image})$ important?

A: It captures: Generative caption/VQA factorization. Use it as a compact exam sentence, then define each symbol.

Card 52

Q: In plain English, why is $\text{CrossAttn}(Q_{\text{text}}, K_{\text{img}}, V_{\text{img}})$ important?

A: It captures: Fusion mechanism. Use it as a compact exam sentence, then define each symbol.

Card 53

Q: In plain English, why is $v=f_{\text{img}}(\text{image})$, $t=f_{\text{txt}}(\text{text})$ important?

A: It captures: Image and text embeddings. Use it as a compact exam sentence, then define each symbol.

Card 54

Q: In plain English, why is $s(\text{image}, \text{text})=v^T t / \tau$ important?

A: It captures: Temperature-scaled similarity. Use it as a compact exam sentence, then define each symbol.

Card 55

Q: In plain English, why is $P(\text{text}_i | \text{image}_i) = \exp(s_{ii}) / \sum_j \exp(s_{ij})$ important?

A: It captures: Contrastive image-to-text probability. Use it as a compact exam sentence, then define each symbol.

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Explanation

Chapter 8: Vision-Language Models

How this section is organized

Each item below is a heading taken from the corresponding Jupyter notebook in the same order it appears there. The explanation beside it tells you what that notebook section is trying to teach and why it matters for the review.

Source notebook

Chapter_8_Vision_Language_Models.ipynb

Headings included

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Notebook headings with explanation

1. Chapter 8 Vision-Language Models

This notebook heading introduces chapter 8 vision-language models and explains how it supports the chapter concepts, equations, method steps, or implementation workflow.

2. Learning Objectives

After completing this chapter, you should be able to: 1. Explain the two dominant VLM architecture families: - Dual-encoder contrastive (CLIP-like) and why it enables *retrieval* and *zero-shot classification*. - Fusion / encoder-decoder (cross-attention + generation) and why it enables *captioning, VQA, instruction following*.

3. What are Vision-Language Models?

Vision-Language Models (VLMs) learn to connect visual semantics (objects, actions, relations) with linguistic semantics (words, phrases, sentences). They support tasks where *either modality can query the other*.

4. Two dominant architecture families

Dual-encoder (contrastive) VLMs — *CLIP-like* Two separate encoders map each modality to a shared embedding space: - Image encoder: $f_{\theta}(x)$ in $\mathbb{R}^d$ - Text encoder: $g_{\phi}(y)$ in $\mathbb{R}^d$ Typically we L2-normalize: $v = f_{\theta}(x) / \|f_{\theta}(x)\|$, $t = g_{\phi}(y) / \|g_{\phi}(y)\|$ Similarity (cosine similarity with temperature): $s(x,y) = \frac{v \cdot t}{\tau}$ Where τ is the temperature. Strengths - Extremely efficient retrieval (cache embeddings; use ANN search) - Zero-shot classification via prompt embeddings -

5. Simple CLIP Implementation

To make CLIP concrete, we will implement a small CLIP-style dual encoder and train it on MNIST digits paired with synthetic captions like: - “a photo of the digit 7” - “handwritten number seven” This is not intended to match real CLIP performance but an example that. - builds the similarity matrix s_{ij} - applies the symmetric contrastive loss - demonstrates zero-shot classification and retrieval > If MNIST download fails (network restrictions), you can point root to a local cached dataset. A tiny. The notebook also includes 11 nearby code cells for implementation, demonstration, or visualization of this heading.

6. Fusion-based encoder-decoder VLM on MNIST

A fusion / encoder-decoder VLM differs from CLIP-like dual encoders in one key way: - CLIP-like (dual encoder): produces *separate* embeddings for image and text and aligns them via contrastive learning. - Best for retrieval and zero-shot classification (embedding-space decision rules). - Fusion / encoder-decoder: fuses modalities via cross-attention and learns to generate text given an image. - Best for captioning, VQA, and instruction following. Objective (image-conditioned language modeling) Given an. The notebook also includes 4 nearby code cells for implementation, demonstration, or visualization of this heading.

7. CLIP and Fusion as Generative Models

CLIP CLIP is not itself a generative model. Its purpose is to learn a shared representation space for images and text so that semantically matching image-text pairs have similar embeddings.

8. Beyond CLIP: key variants and why they differ	ALIGN - Similar dual-encoder recipe - Emphasizes training at scale with noisy alt-text pairs LiT (Locked-image Tuning) - Lock a strong image encoder (often pretrained on classification) - Tune text side for alignment - Can yield strong transfer SigLIP CLIP uses a batch-softmax normalization. SigLIP replaces it with a pairwise sigmoid objective (conceptually): For a pair with label $y_{ij} \in \{+1, -1\}$, $L_{\text{sig}} = \sum_{i,j} \log \left(1 + \exp(-y_{ij} s_{ij}) \right)$.
9. Full, runnable pretrained CLIP pipeline	Below are two complete “batteries-included” options to use an existing CLIP implementation: 1. Hugging Face transformers (CLIPModel + CLIPProcessor) 2. The notebook also includes 10 nearby code cells for implementation, demonstration, or visualization of this heading.
10. Small / efficient CLIP (TinyCLIP, MobileCLIP):	In practice, many VLM deployments want <i>*fast*</i> embeddings on edge devices (phones, embedded GPUs/NPUs). This creates an accuracy-latency-memory tradeoff. The notebook also includes 14 nearby code cells for implementation, demonstration, or visualization of this heading.
11. Extensions beyond CLIP (what people built on top of CLIP-like encoders)	CLIP’s core contribution is a transferable image-text embedding space. A large portion of VLM research since 2021 can be viewed as <i>*how to extend this interface*</i> for: - Better training/scaling (losses, data curation, multilinguality) - Adaptation (prompt learning, adapters, few-shot) - Dense prediction (segmentation) and localization (detection / grounding) - Reasoning with multimodal LLMs Below is a structured map with references.